



Solving Trigonometric Equations Using Identities

Trigonometry Worksheet · Grade 10-12 · numberbender.com

Name: _____

Date: _____

Score: / 9

Learning Objectives

- Apply Pythagorean identities to simplify trigonometric equations
- Solve trigonometric equations on the interval $[0, 2\pi)$
- Factor trigonometric expressions to find all solutions

Solve each trigonometric equation for θ on the interval $[0, 2\pi)$, showing all identity substitutions and algebraic steps.

1. Solve for θ on $[0, 2\pi)$.

$$\sin^2 \theta - \cos^2 \theta = 0$$

Answer: _____

2. Solve for x on $[0, 2\pi)$.

$$\cos^2 x + \cos x - \sin^2 x = 0$$

Answer: _____

3. Solve for θ on $[0, 2\pi)$.

$$2\sin^2 \theta - 1 = 0$$

Answer: _____

4. Solve for θ on $[0, 2\pi)$.

$$2\cos^2 \theta - \cos \theta - 1 = 0$$

Answer: _____

5. Solve for θ on $[0, 2\pi)$.

$$\sin^2 \theta + \cos \theta = 1$$

Answer: _____

6. Solve for x on $[0, 2\pi)$.

$$2\cos^2 x + 3\cos x + 1 = 0$$

Answer: _____

7. Solve for θ on $[0, 2\pi)$.

$$\tan^2 \theta - 1 = 0$$

Answer: _____



Math is hard. So we made it human.
Labanan ang agwat.

Free to print & share (CC BY 4.0)

Solving Trig Equations Using Identities Worksheet

More free worksheets, video lessons & courses:

numberbender.com/math-worksheets

Page 1 of 4 · Video: youtu.be/bHAEjYJu5cE



Watch the lesson



More free sheets

8. Solve for θ on $[0, 2\pi)$.

$$2\sin^2 \theta + \sin \theta - 1 = 0$$

Answer: _____

9. Solve for x on $[0, 2\pi)$.

$$\cos^2 x - \sin^2 x = \frac{1}{2}$$

Answer: _____





Encourage students to treat θ like a variable x and remind them to include $\pm$ when taking square roots and to rationalize radical denominators.

Solutions

1. Solve for θ on $[0, 2\pi)$.

$$\sin^2 \theta - \cos^2 \theta = 0$$

- Replace sine squared theta with one minus cosine squared theta using the Pythagorean identity.
- Combine like terms to get one minus two cosine squared theta equals zero.
- Subtract one from both sides and divide by negative two to isolate cosine squared theta.
- Take the square root of both sides, remembering plus or minus, then rationalize the denominator.
- Use the unit circle to find the four angles where cosine theta equals plus or minus root two over two.

Answer: $\theta = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}$

2. Solve for x on $[0, 2\pi)$.

$$\cos^2 x + \cos x - \sin^2 x = 0$$

- Replace sine squared x with one minus cosine squared x to make every term contain cosine.
- Combine like terms to obtain two cosine squared x plus cosine x minus one equals zero.
- Factor the quadratic in cosine x as two cosine x minus one times cosine x plus one.
- Set each factor equal to zero to get cosine x equals one half or cosine x equals negative one.
- Use the unit circle to identify the angles π over three, five π over three, and π .

Answer: $x = \frac{\pi}{3}, \pi, \frac{5\pi}{3}$

3. Solve for θ on $[0, 2\pi)$.

$$2\sin^2 \theta - 1 = 0$$

- Add one to both sides and divide by two to isolate sine squared theta.
- Take the square root of both sides and include the plus or minus sign.
- Rationalize the denominator to get sine theta equals plus or minus root two over two.
- Use the unit circle to list the four quadrantal solutions in $[0, 2\pi)$.

Answer: $\theta = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}$

4. Solve for θ on $[0, 2\pi)$.

$$2\cos^2 \theta - \cos \theta - 1 = 0$$

- Treat cosine theta as a variable and factor the quadratic as two cosine theta plus one times cosine theta minus one.
- Set each factor equal to zero to get cosine theta equals negative one half or cosine theta equals one.
- Use the unit circle to find theta equals two π over three and four π over three for the first factor.
- Identify theta equals zero as the solution where cosine theta equals one on the interval.

Answer: $\theta = 0, \frac{2\pi}{3}, \frac{4\pi}{3}$



5. Solve for θ on $[0, 2\pi)$.

$$\sin^2 \theta + \cos \theta = 1$$

- Replace sine squared theta with one minus cosine squared theta using the Pythagorean identity.
- Simplify to get negative cosine squared theta plus cosine theta equals zero.
- Factor out cosine theta to get cosine theta times one minus cosine theta equals zero.
- Set each factor equal to zero, giving cosine theta equals zero or cosine theta equals one.
- Use the unit circle to list theta equals zero, pi over two, and three pi over two.

Answer: $\theta = 0, \frac{\pi}{2}, \frac{3\pi}{2}$

6. Solve for x on $[0, 2\pi)$.

$$2\cos^2 x + 3\cos x + 1 = 0$$

- Factor the quadratic in cosine x as two cosine x plus one times cosine x plus one.
- Set each factor equal to zero to get cosine x equals negative one half or cosine x equals negative one.
- Use the unit circle to find x equals two pi over three and four pi over three for the first factor.
- Identify x equals pi as the solution where cosine x equals negative one.

Answer: $x = \pi, \frac{2\pi}{3}, \frac{4\pi}{3}$

7. Solve for θ on $[0, 2\pi)$.

$$\tan^2 \theta - 1 = 0$$

- Add one to both sides to obtain tangent squared theta equals one.
- Take the square root of both sides and include the plus or minus sign.
- Solve tangent theta equals one and tangent theta equals negative one separately.
- Use the unit circle to list the four reference angles at pi over four in each quadrant.

Answer: $\theta = \frac{\pi}{4}, \frac{3\pi}{4}, \frac{5\pi}{4}, \frac{7\pi}{4}$

8. Solve for θ on $[0, 2\pi)$.

$$2\sin^2 \theta + \sin \theta - 1 = 0$$

- Factor the quadratic in sine theta as two sine theta minus one times sine theta plus one.
- Set each factor equal to zero to get sine theta equals one half or sine theta equals negative one.
- Use the unit circle to find theta equals pi over six and five pi over six for the first factor.
- Identify theta equals three pi over two as the solution where sine theta equals negative one.

Answer: $\theta = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{3\pi}{2}$

9. Solve for x on $[0, 2\pi)$.

$$\cos^2 x - \sin^2 x = \frac{1}{2}$$

- Replace sine squared x with one minus cosine squared x using the Pythagorean identity.
- Simplify to get two cosine squared x minus one equals one half.
- Add one to both sides and divide by two to isolate cosine squared x.
- Take the square root of both sides, include plus or minus, and rationalize to get cosine x equals plus or minus root three over two.
- Use the unit circle to list all four solutions in $[0, 2\pi)$.

Answer: $x = \frac{\pi}{6}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{11\pi}{6}$



Math is hard. So we made it human.
Labanan ang agwat.

Free to print & share (CC BY 4.0)

Solving Trig Equations Using Identities Worksheet

More free worksheets, video lessons & courses:

numberbender.com/math-worksheets

Page 4 of 4 · Video: youtu.be/bHAEjYJu5cE



Watch the lesson



More free sheets